



SLTC
Research University

End Semester Project Final Report

Name: Siluni Dulvanya Pasqual

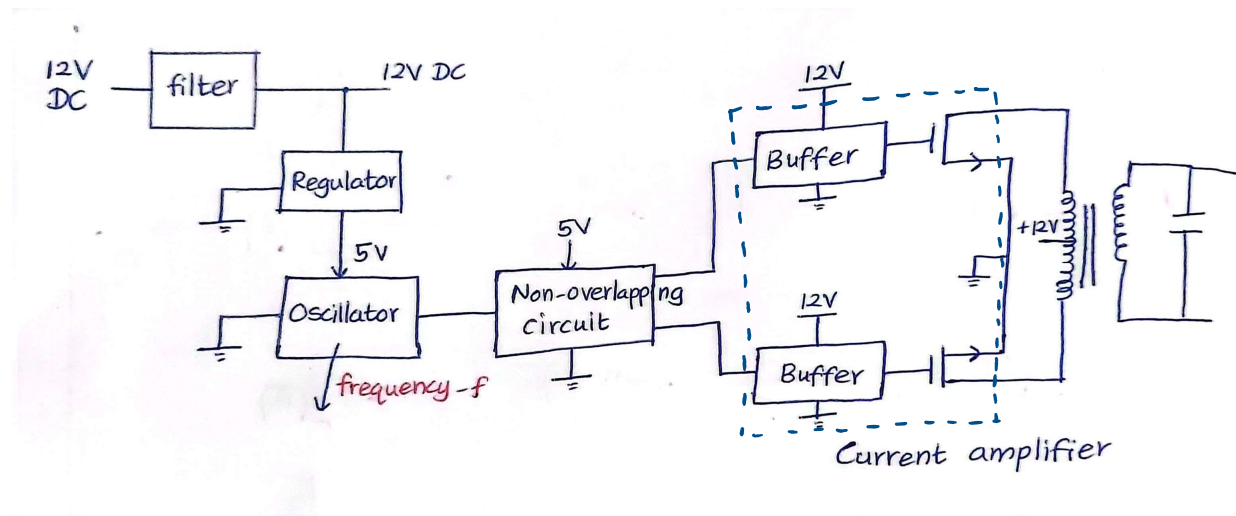
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Module: EEE2100 Analog Electronics Project

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This project entails the creation of a DC to AC converter. The fundamental schematic or methodology to be employed in the design of this project involves the following.



Fundamental schematic which outlines the project.

Pre calculations

- Specifications are based on the last three digits of our index number (XYZ).

Registration Number: 22UG2-0145 $\Rightarrow$ X = 1, Y = 4, Z = 5

- $V_{in} = 12V \pm 0.Z \text{ Volts} = 12V \pm 0.5 = 11.5V \text{ to } 12.5V$
- $V_{out}^{rms} = (20 + YZ) \text{ Volts} = 20 + 45 = 65V$
- $I_{out}^{max} = \left(\frac{Z}{2}\right) \times 100mA = 250mA$
- $f - \text{frequency} = Y.ZkHz = 4.5kHz$
- Switching noise of input source = $ZYkHz = 54kHz$
- Input voltage ripple = $10 \times ZmV = 10 \times 5mV = 50mV$

For the **first task**, we were asked to design a voltage regulator which gives a 5V output to a 12V input given. And, to design a suitable filter to avoid the ripple and give a clear DC input if there exists a small ripple in the input signal given from the DC power supply.

Here are some reasons why voltage ripples can be applied in a DC input.

- Power supply imperfections** – In the practical world, the power supply may not provide a perfectly constant voltage levels due to inherent imperfections.
- Load variations** – Changes in the connected load, or sudden variations in the current drawn through the load can cause voltage ripples.
- Noise and interference** – External factors such as electromagnetic interference (EMI) or radio frequency interference (RFI) can infiltrate the DC power lines and contribute to voltage fluctuations.

1. Design Process of the Filter Circuit

As in our pre calculations, the switching noise of the input power source is 54kHz. For this circuit's requirements, we are interested in the frequencies below the cut-off frequency. So, we need to allow low frequencies to pass through while attenuating high frequencies. For that, a Low Pass Filter would be suitable.

For this, I chose a LC Low Pass Filter and here are the reasons for choosing that upon RC filters and RL filters.

- **Lower Bandpass ripple**

LC filters provide superior ripple rejection compared to RC filters. The combination of an inductor and capacitor allows for more effective filtering of both high and low-frequency components, resulting in a cleaner DC output.

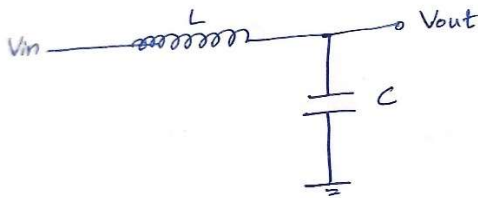
- **Highly efficient at low frequencies**

For low-frequency noise and fluctuations, inductors (L) in LC filters are more effective than resistors (R) in RC filters. Inductors resist changes in current, contributing to better low-frequency filtering.

- **Reduced Power Dissipation**

LC filters typically have lower power dissipation compared to RC filters.

This is the simple design of the LC circuit. The output frequency in a filter is the point where the output power is halved or exhibits a -3dB (-6dB attenuation).



$$f_{cut-o} = 4.5kHz$$

$$f_{cut-o} = \frac{1}{2\pi\sqrt{LC}}$$

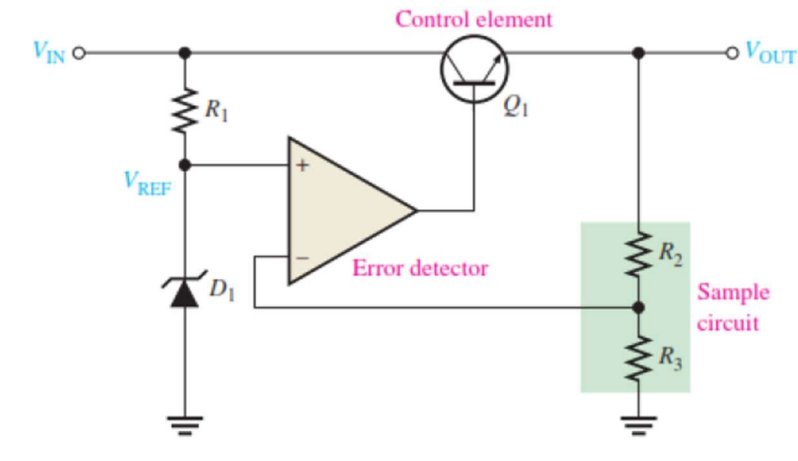
Substituting values to this, $LC = 1.2508 \times 10^{-9}$

$L = 125\mu H$, $C = 0.01\mu F$

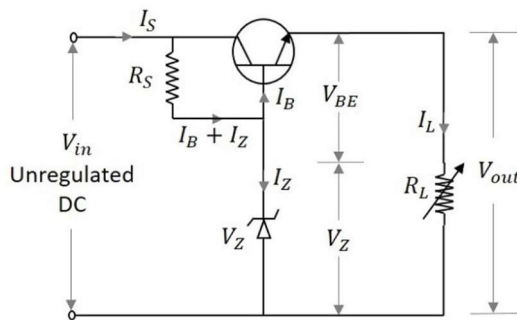
2. Design Process of the Voltage Regulator

Here's the basic circuit design chosen for the voltage regulator circuit.

Basic op-amp series regulator



2.1 Specifications for the design



Design A

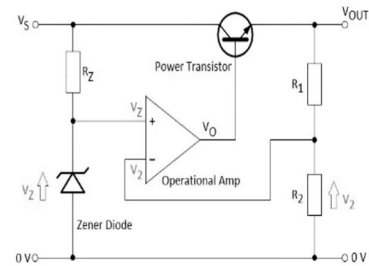


Fig. 1. Voltage regulator using operational amplifier

Design B

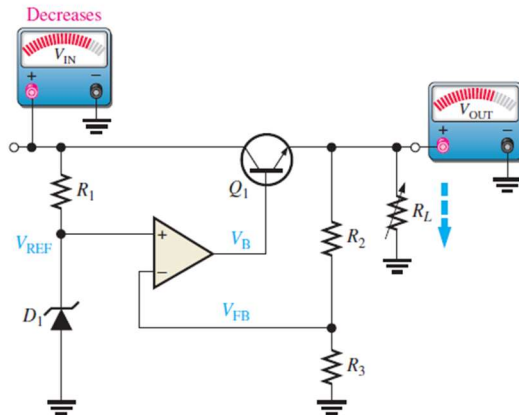
The op-amp in the Design B provides additional feedback and amplification, allowing for better regulation and control of the output voltage, resulting in enhanced accuracy and reduced sensitivity to variations in load and input voltage.

The basic operation of the op-amp series regulator is as follows.

The series regulator, outlined in Figure below, operates by using resistors R_2 and R_3 to monitor changes in the output voltage. When the output decreases due to factors like a drop in V_{IN} or an increase in I_L from reduced R_L , the voltage drop is detected by the op-amp's inverting input via the voltage divider. This prompts an increase in the op-amp's output, V_B , which, in turn, boosts the emitter voltage, V_{OUT} , until equilibrium is restored. The regulator ensures a constant output voltage despite variations. Conversely, if the output attempts to increase, a similar process

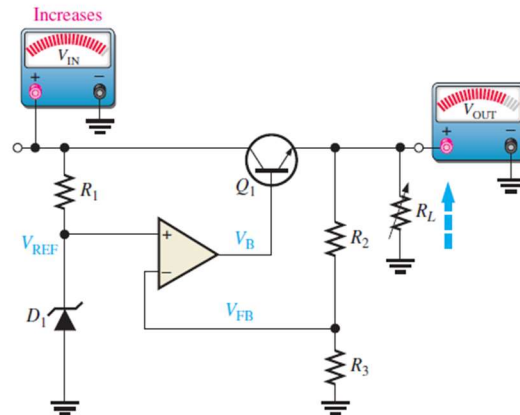
occurs. Notably, the op-amp functions as a **noninverting amplifier**, with V_{REF} as the input, and the $R_2 > R_3$ voltage divider providing negative feedback. The closed-loop voltage gain is,

$$A_{cl} = 1 + \frac{R_2}{R_3}$$



(a) When V_{IN} or R_L decreases, V_{OUT} drops slightly. The slight drop is sampled by the feedback voltage, V_{FB} , and the op-amp immediately increases V_B . This increase causes the output to remain nearly the same as before the original drop. Changes in V_{OUT} are exaggerated for illustration.

When V_{IN} (or R_L) stabilizes at its new lower value, the voltages turn to their original values, thus keeping V_{OUT} nearly constant as a result of the negative feedback.



(b) When V_{IN} or R_L increases, V_{OUT} increases slightly. The slight increase is sampled by the feedback voltage, V_{FB} , and the op-amp immediately decreases V_B . This decrease causes the output to remain nearly the same as before the original increase.

When V_{IN} (or R_L) stabilizes at its new higher value, the voltages return to their original values, thus keeping V_{OUT} nearly constant as a result of the negative feedback.

2.2 Explanation of the circuit

The fundamental idea involves employing feedback to adjust the output voltage in response to alterations in the load or input voltage. The process is detailed as follows:

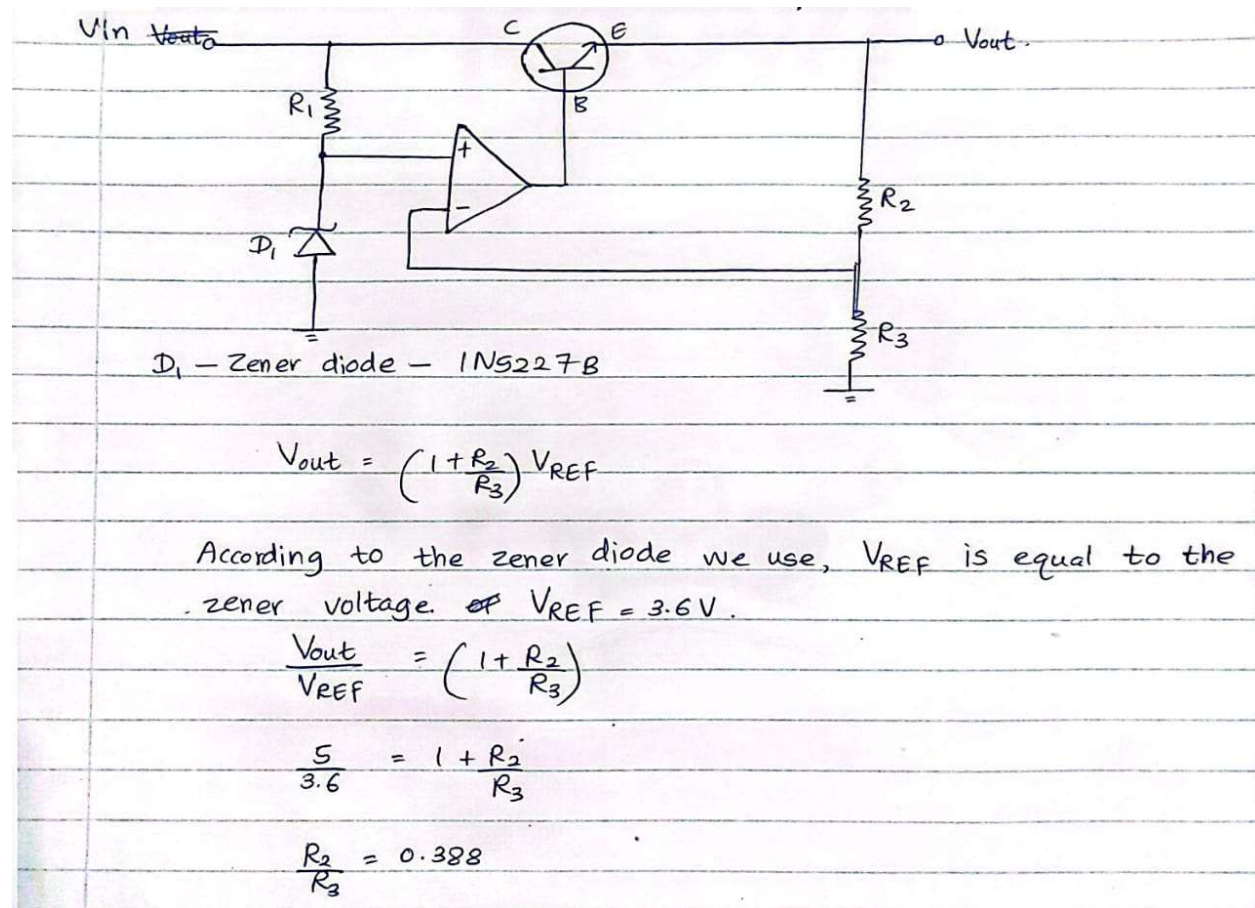
Reference Voltage: The regulator incorporates a stable reference voltage, typically generated using a Zener diode. This reference voltage serves as a constant benchmark for the op-amp.

Feedback Loop: Ensuring stability, a crucial feedback loop is established. It detects any deviation in the output voltage and ensures necessary adjustments. Depending on the application, the response within the loop may be either positive or negative.

Voltage Regulation: Achieving a consistent output voltage involves the op-amp continuously comparing the actual output voltage with the reference value and adjusting the series pass element (BJT). This dynamic, real-time process responds to variations in input voltage or load conditions.

2.3 Design calculations based on the specifications

This is the design with properly calculated values and properly choose for the voltage regulator circuit.



For R_2 and R_3 I chose 390Ω and $1k\Omega$.

Zener biasing current = $20mA$.

Same current goes through R_1 as well

$$\frac{V_{in} - V_{REF}}{20mA} = R_1$$

For

$$\frac{12 - 3.6}{20 \times 10^{-3}} = R_1$$

For

$$R_1 = 420\Omega$$

For R_1 I chose a 470Ω resistor.

2.3 Component Selection

(A) BJT transistor selection

The $I_{\max}$ for the circuit design is **250mA**. For this voltage regulator circuit, the current going through is nearly 0.08A (when a load is connected). So, the BJT transistor which is chosen should be able to handle that current. So, a BJT transistor which has its maximum current rating of I_c , more than 0.08A is chosen.

BJT selected – **BC547** npn transistor

Datasheet: <https://www.sparkfun.com/datasheets/Components/BC546.pdf>

(B) Zener diode selection

A Zener which has its' Zener voltage less than 5V should be selected, as the output voltage is 5V, and according to the circuit, V_{REF} should be less than 5V. Also, this Zener diode should be able to handle the maximum current in an emergency situation.

Zener diode selected – **3.6V(1/2W) Zener diode**

Zener voltage = 3.6V

Datasheet: <https://circuits-diy.com/wp-content/uploads/2021/05/1N747-1.pdf>

(C) Op-amp selection

An op-amp is needed which can handle 250mA and give an amplified output of doubled the input voltage. The availability of the components and the cost is also considered.

LM741/UA741 op-amp was chosen.

Datasheet: <https://pdf1.alldatasheet.com/datasheet-pdf/view/25555/STMICROELECTRONICS/UA741.html>

2.4 Effect of component tolerance

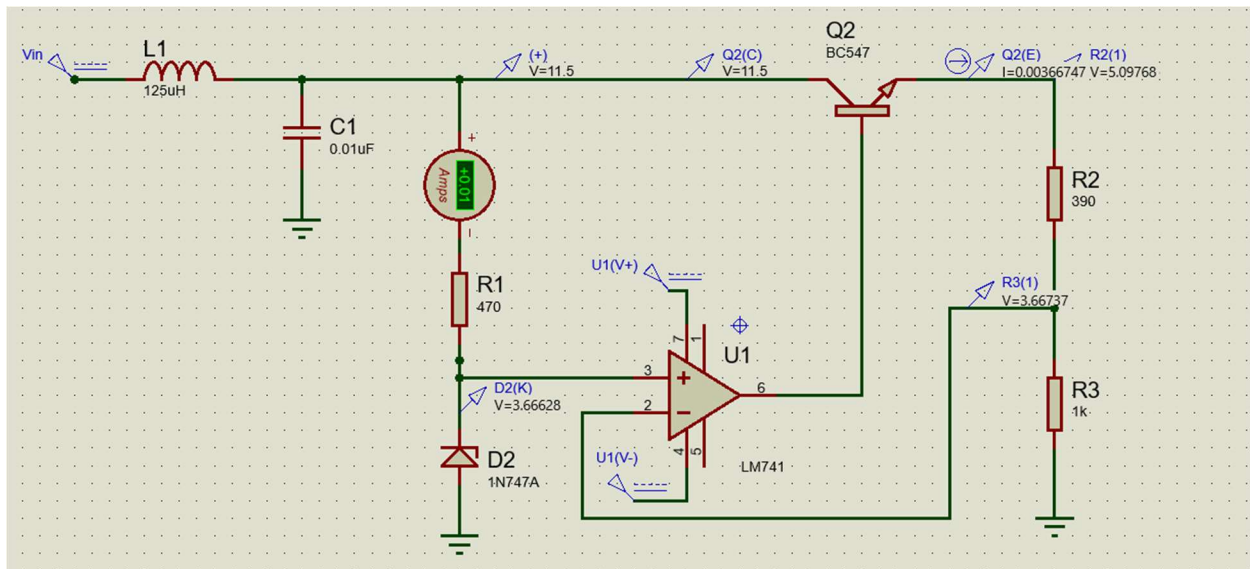
(a) Zener voltage

There is a 5% tolerance in V_z , so the Zener voltage is in between 3.42V-3.78V.

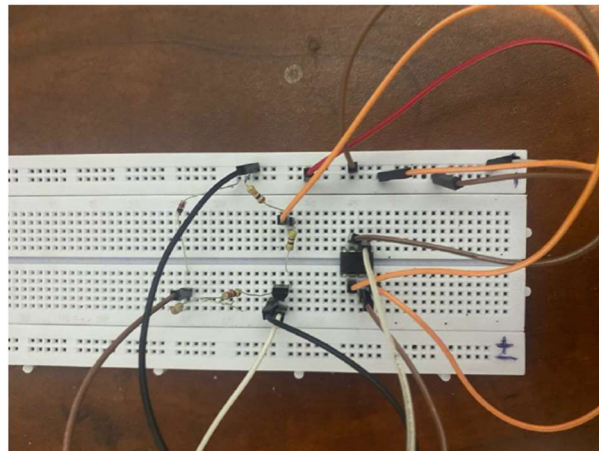
(b) Resistor values

Normally there is a 5% tolerance in the available E12 series resistors.

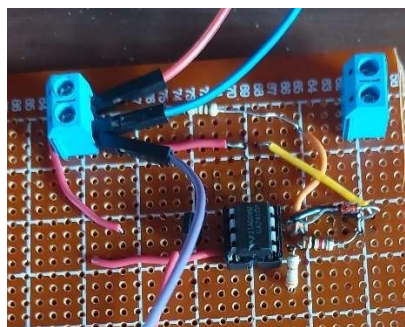
This is the circuit which was designed using Proteus 8 simulation software. (Including both the filter circuit and the regulator circuit)



This is the circuit which was made and tested in the lab.



This is the same circuit soldered in the dot board.



Output voltage of the voltage regulator circuit

Output voltage = 5.4V



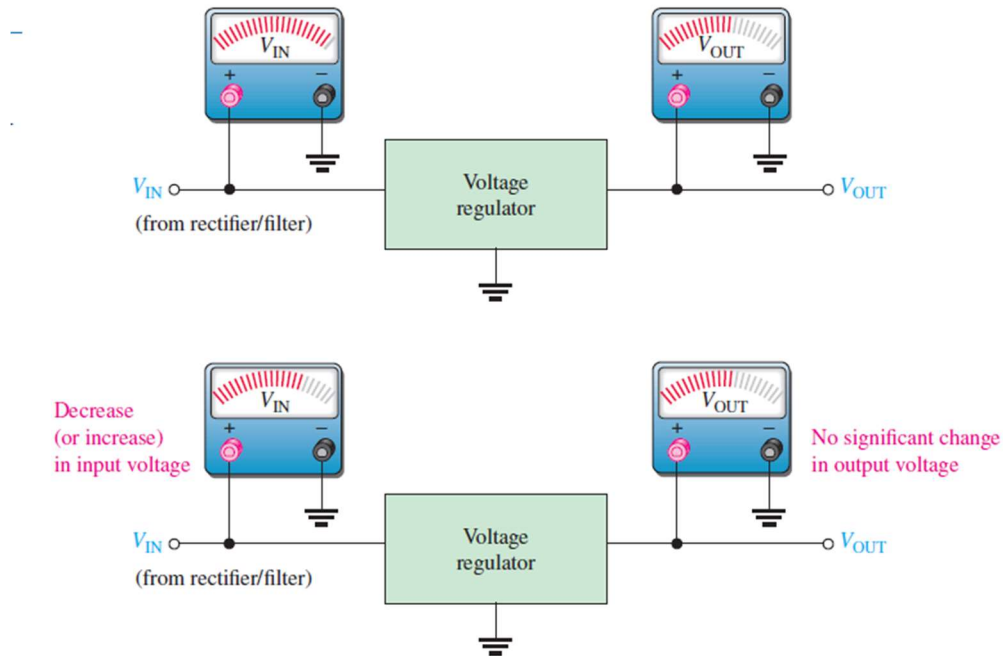
❖ **Why can't we directly use 12V DC and why do we need to regulate that voltage to 5V?**

Changing the initial 12V power to 5V in the DC to AC converter is like making sure all the parts can handle the right amount of power without getting damaged. It also helps the circuit work more efficiently by using less power and reduces the chances of electrical noise messing things up. Using 5V matches up with how digital parts like to work, making them play well together. Overall, it helps the circuit be more reliable and work consistently.

2.5 Testing the circuit

2.5.1 Line regulation

$$\text{Line regulation} = \frac{(\Delta V_{\text{OUT}}/V_{\text{OUT}})100\%}{\Delta V_{\text{IN}}}$$



- When $V_{\text{in}}:\text{min}$

$$\text{Simulated Load regulation} = \frac{0.002}{0.5} \times 100\% = 0.4\%$$

$$\text{Tested line regulation} = \frac{0.01}{0.5} \times 100\% = 2\%$$

- When $V_{\text{in}}:\text{mid}$

$$\text{Simulated Line regulation} = 0$$

$$\text{Tested Line regulation} = 0$$

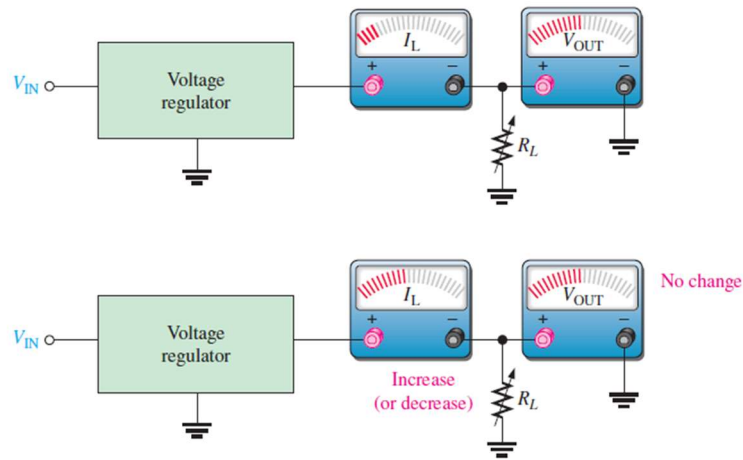
- When $V_{\text{in}}:\text{max}$

$$\text{Simulated Line regulation} = \frac{3.7 \times 10^{-3}}{0.5} \times 100\% = 0.74\%$$

$$\text{Tested Line Regulation} = \frac{0.002}{0.5} \times 100\% = 0.4\%$$

2.5.2 Load regulation

$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}} \right) 100\%$$



Calculations for the load resistance in each case

- When $I=0$
Load regulation = 0

- When $I=\text{max}/2$

$$\text{Simulated Load regulation} = \frac{(5.031 - 5.0288)}{5.031} \times 100\% = 0.04\%$$

$$\text{Tested Line Regulation} = \frac{(5.56 - 5.42)}{5.56} \times 100\% = 2.51\%$$

- When $I=\text{max}$

$$\text{Simulated Load regulation} = \frac{(5.0314 - 5.0317)}{5.0314} \times 100\% = -5.962 \times 10^{-3}\%$$

$$\text{Tested Line Regulation} = \frac{(5.78 - 5.3)}{5.78} \times 100\% = 8.3\%$$

2.5.3 Efficiency

$$\text{Efficiency}(\%) = \frac{P_{in}}{P_{out}} \times 100\%$$

- When $I=0 \Rightarrow$ Can't calculate.
- When $I = \text{max}/2$

$$I_B \approx 1\text{mA}, \text{ So } I_E = I_C = 80\text{mA}$$

$$\text{Efficiency}(\text{simulated}) = \frac{5}{12} \times 100\% = 41.667\%$$

$$\text{Tested Efficiency} = \frac{5.37}{12} \times 100\% = 44.75\%$$

- When $I = \text{max}$

$$\text{Efficiency}(\text{simulated}) = \frac{5}{12} \times 100\% = 41.667\%$$

$$\text{Tested Efficiency} = \frac{5.4}{12} \times 100\% = 45\%$$

2.6 Results and Analysis

Comparison between the test reading and the simulation

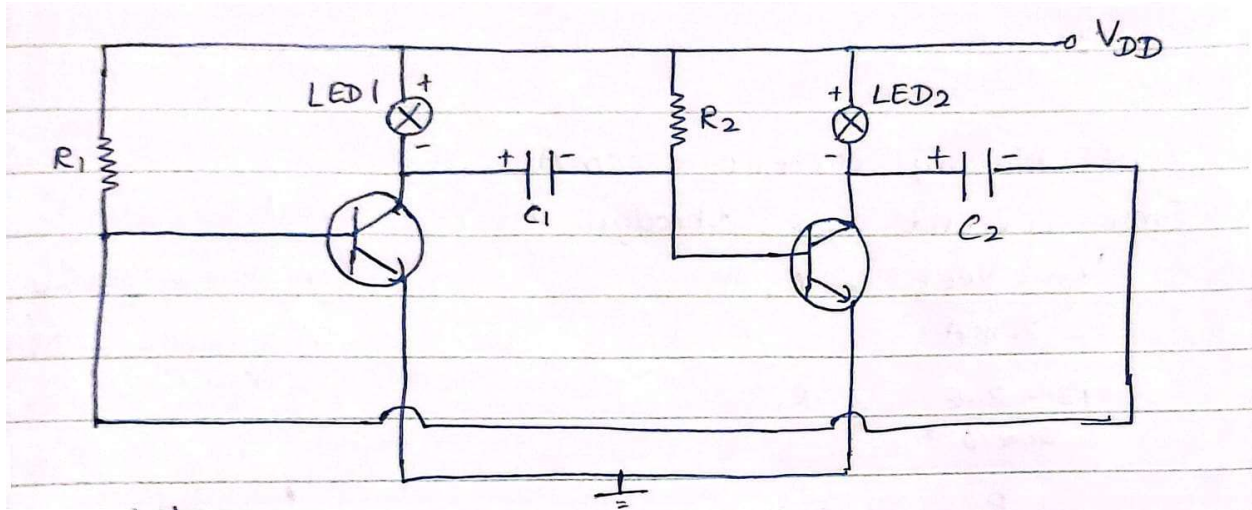
		Test Reading	Simulation result
Line regulation	Vin:min	2%	0.4%
	Vin:mid	0%	0%
	Vin:max	0.4%	0.74%
Load regulation	I=0	-	-
	I=max/2	0.04%	2.51%
	I=max	$-5.962 \times 10^{-3}\%$	8.3%
Efficiency	I=0	-	-
	I=max/2	41.667%	44.75%
	I=max	41.667%	45%

3. References for Task-01

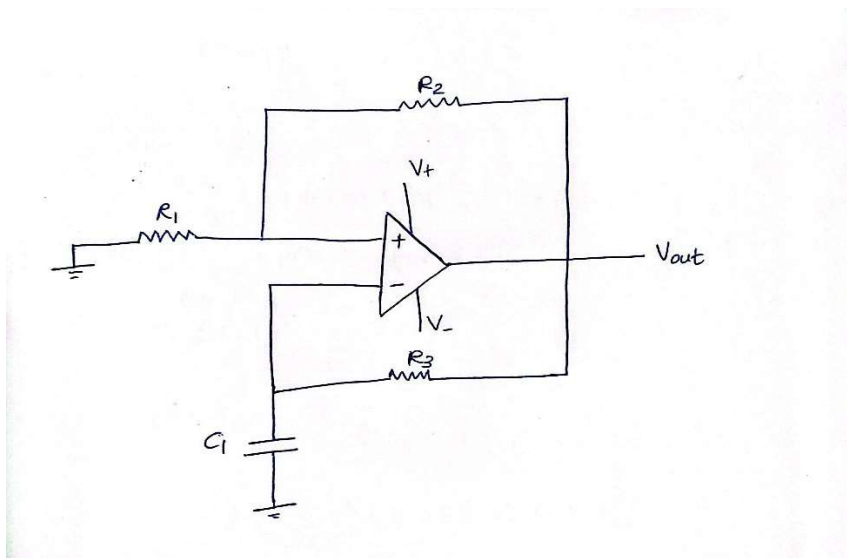
- (Floyd)
- <https://ae-iitr.vlabs.ac.in/exp/voltage-regulator/theory.html#:~:text=A%20voltage%20regulator%20circuit%20using,a%20large%20range%20of%20voltages.>

4. Designing the Oscillator

4.1 Circuit diagram



Astable multivibrator using transistors.



Opamp oscillator circuit

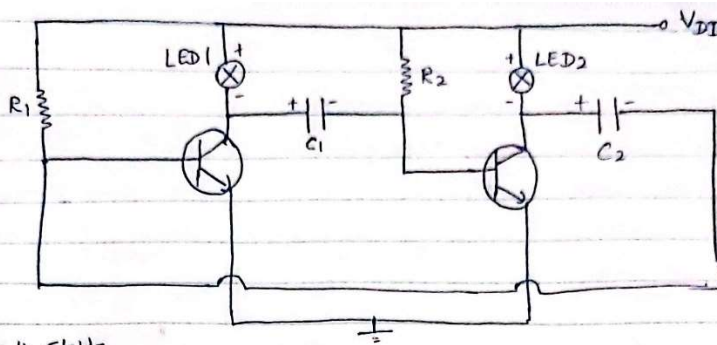
From these two designs, first, the astable multivibrator using transistors was made. That circuit was chosen considering these,

- The needed components cost less than using Op-amps and ICs.
- The needed oscillating frequency can be decided by us.
- We can directly use the regulated voltage coming from the voltage regulator to the power line of the oscillator without giving power from a separate power supply.

4.2 Circuit operation

1. The circuit starts in an unstable state where one transistor is conducting (ON), and the other is off (OFF).
2. Initially, the conducting transistor (Q1) allows current to flow through resistor R1 and charge the capacitor C1.
3. As the capacitor C1 charges, the voltage across it gradually increases.
4. When the voltage across the charging capacitor C1 reaches a certain threshold, it triggers the non-conducting transistor (Q2) to switch ON.
5. With Q2 ON, the capacitor C2 discharges through the resistor R2, causing the voltage across C2 to drop.
6. When the voltage falls below a certain threshold, it turns off Q2, and the cycle repeats.
7. The process alternates between Q1 conducting and charging capacitor C1 and Q2 conducting and discharging capacitor C2.
8. This continuous switching between the two transistors results in an output square wave with a specific frequency determined by the values of the resistors R1 and R2 and capacitors C1 and C2 in the circuit.

4.3 Calculations



$$f = 4.5 \text{ kHz}$$

Calculations for this design,

$$V_C(t) = V_{DD}(\infty) + [V_{DD}(0) - V_{DD}(\infty)]e^{-t/RC}$$

$$V_C(0) = -V_{DD} + V_{BE}$$

$$V_{DD}(\infty) = V_{DD}$$

$$t = T_1 \Rightarrow V_C(T_1) = V_{BE}$$

$$V_{BE} = V_{DD} + [-V_{DD} + V_{BE} - V_{DD}]e^{-t/RC}$$

$$e^{-t/RC} = \frac{V_{BE} - V_{DD}}{V_{BE} - 2V_{DD}}$$

$$\begin{aligned} V_{DD} > V_{BE} \\ e^{-t/RC} &= \frac{-V_{DD}}{-2V_{DD}} = \frac{1}{2} \end{aligned}$$

$$T_1 = -RC \ln\left(\frac{1}{2}\right)$$

$$T_1 = RC \ln(2)$$

$$T = 2T_1 = 2RC \ln(2)$$

$$T = \frac{1}{f} = \frac{1}{4.5 \times 10^3} = 2.2222 \times 10^{-4}$$

$$2.222 \times 10^{-4} = 2RC \ln(2)$$

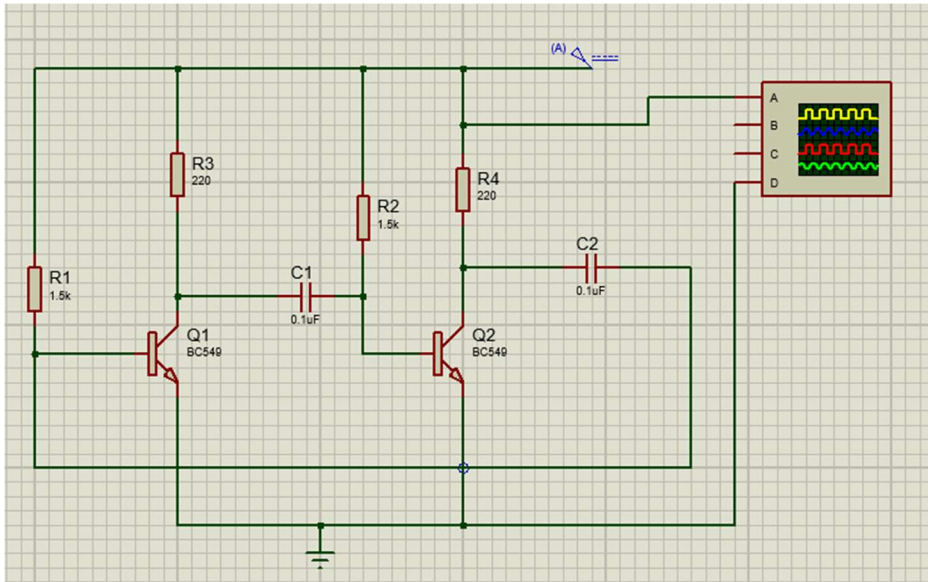
$$RC = 1.6029 \times 10^{-4}$$

The values chosen for R and C are $R=1.5\text{k}\Omega$ and $C=0.1\mu\text{F}$. Here, we use LEDs for the visualization purpose of the circuit. After testing whether the circuit is working, I replaced those two LEDs with two load resistors of 220Ω each.

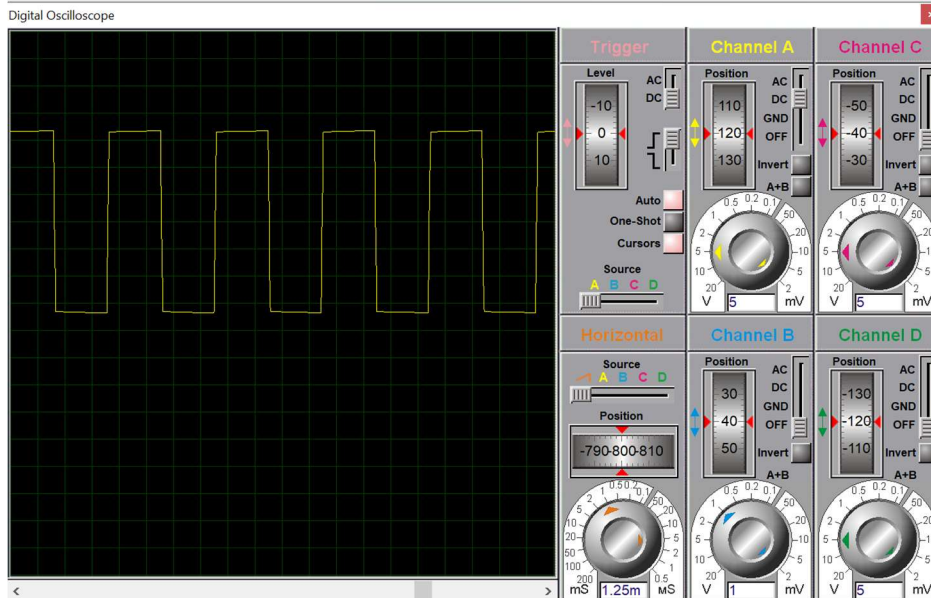
- The exact value of the base resistor can be decided by replacing the resistor with a variable resistor.

4.4 Simulation result

- Simulated circuit

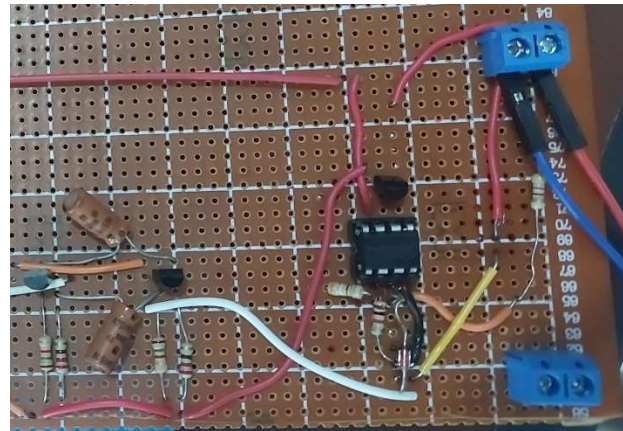
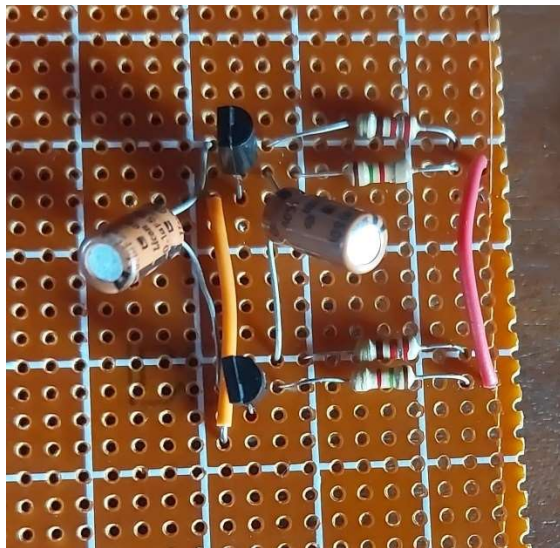


- Output waveform



4.5 Experimental result

This is the circuit made and the output waveform.



Frequency = 4.31kHz

- The reason for this output waveform not being an exact clock signal might be the charging time of the capacitor is higher than the discharging time of the capacitor. That can be omitted by changing the resistor and capacitor values accordingly but keeping the same RC ratio.

When comparing the simulated circuit output waveform and the tested circuit output waveform, the frequencies were differentiated by a factor of nearly 10. The disparity may arise from variations in the equations used by the Proteus simulation software, compared to the equations used to do the calculations.

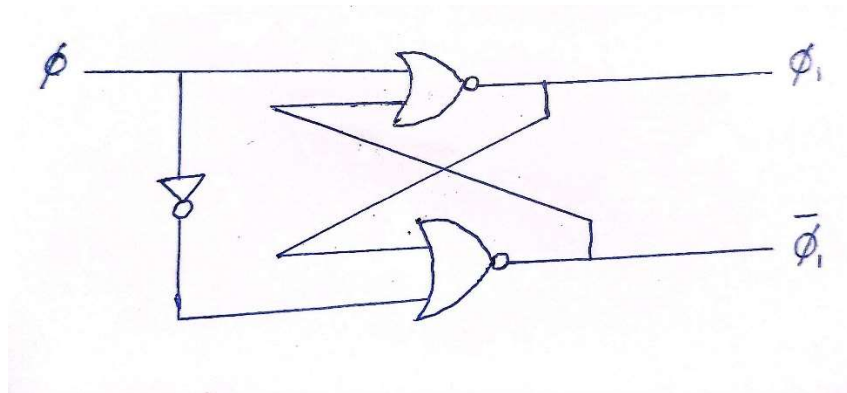
5. Designing the Non-Overlapping-Circuit (NOLC)

5.1 Need of this block for the overall operation of the project

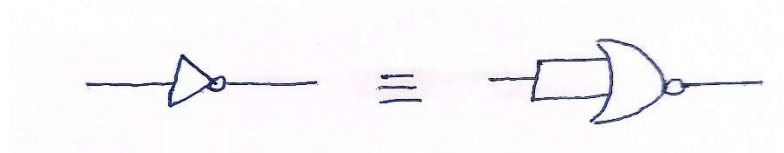
A non-overlapping circuit is crucial for inverters because it prevents a situation where both the input and output signals are active at the same time. This simultaneous activation, if not properly managed, can lead to issues like short circuits and excessive current flow. The non-overlapping circuit introduces a delay between turning off one signal and turning on the other, ensuring a smooth and controlled transition between the two states. This helps prevent harmful effects like overheating and damage to the inverter, ensuring its reliable and efficient operation.

5.2 Operation of NOLC

The Non-Overlapping Clock (NOLC) circuit serves a critical role in digital circuitry by producing two complementary clock signals, CLK and $\sim$ CLK. What sets NOLC apart is its ability to introduce a controlled delay between the rising or falling edges of these signals, preventing any overlap. This feature is essential for ensuring accurate timing and preventing potential race conditions in complex digital systems. The NOLC design is particularly valuable in applications where synchronous circuitry and precise flip-flop control are paramount, contributing to the overall reliability and performance of digital circuits.

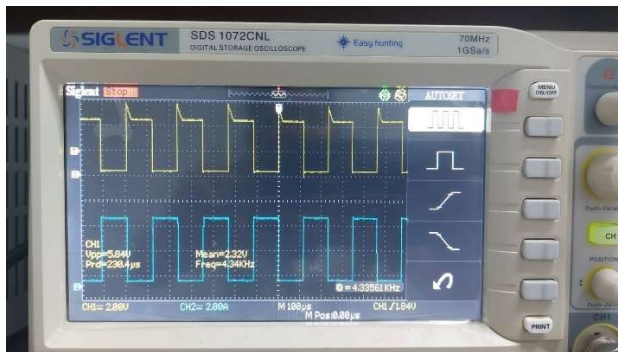
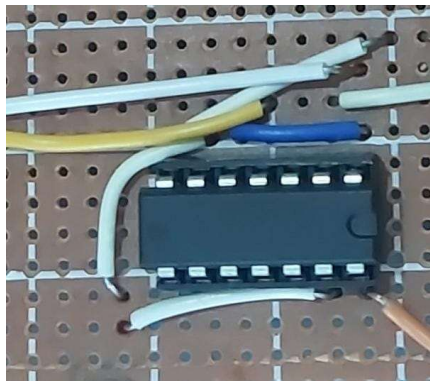


Here, for the NOT gate, I used a NOR gate, as the 7402 NOR gate has 4 gates inside it so we can build this circuit by only using one single NOR gate.



$$\begin{aligned}
 \text{Total time delay} &= \text{Signal period } \left(\frac{1}{f}\right) - \text{Overlap avoidance} + \text{Active duration of one signal} \\
 &= \frac{1}{f} - \text{Min}(D_1, D_2)T + D_1T \\
 &= \frac{1}{4500} - \left(3 \times 10^{-9} \times \frac{1}{4500}\right) + \left(11 \times 10^{-9} \times \frac{1}{4500}\right) \\
 &= \frac{1}{4500} \left(1 - 8 \times 10^{-9}\right) \\
 &\approx 222.22 \mu\text{s}
 \end{aligned}$$

5.3 Experimentally built circuit and output waveform



6 Designing the buffer circuit

6.1 Job of the buffer circuit block

In the inverter design, the buffer circuit plays a pivotal role in bridging the gap between the non-overlapping circuit and the transformer. Its primary functions involve **isolating the output of the non-overlapping circuit from the load, enhancing the signal integrity, and driving the MOSFETs or BJTs with sufficient strength to control the switching of the power devices**. The buffer also aids in **maintaining impedance matching** between different circuit stages, ensuring efficient power transfer. Moreover, it contributes to **waveform shaping, minimizing distortions, and enhancing the stability of the AC output signal**. Overall, the buffer circuit is crucial for maintaining signal quality and reliability throughout the DC to AC conversion process.

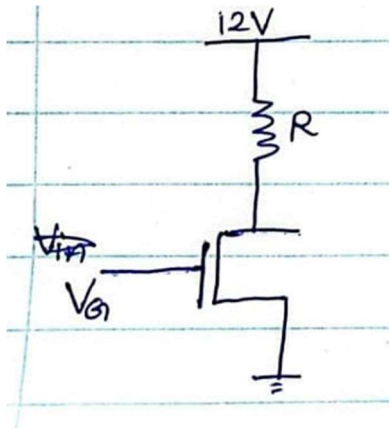
Components used to match the current rating of the circuit.

- D400 NPN transistor
- 470 Ω resistor

D400 transistor is used as it can handle the maximum current rating of the circuit and can handle 5V as V_{BE} .

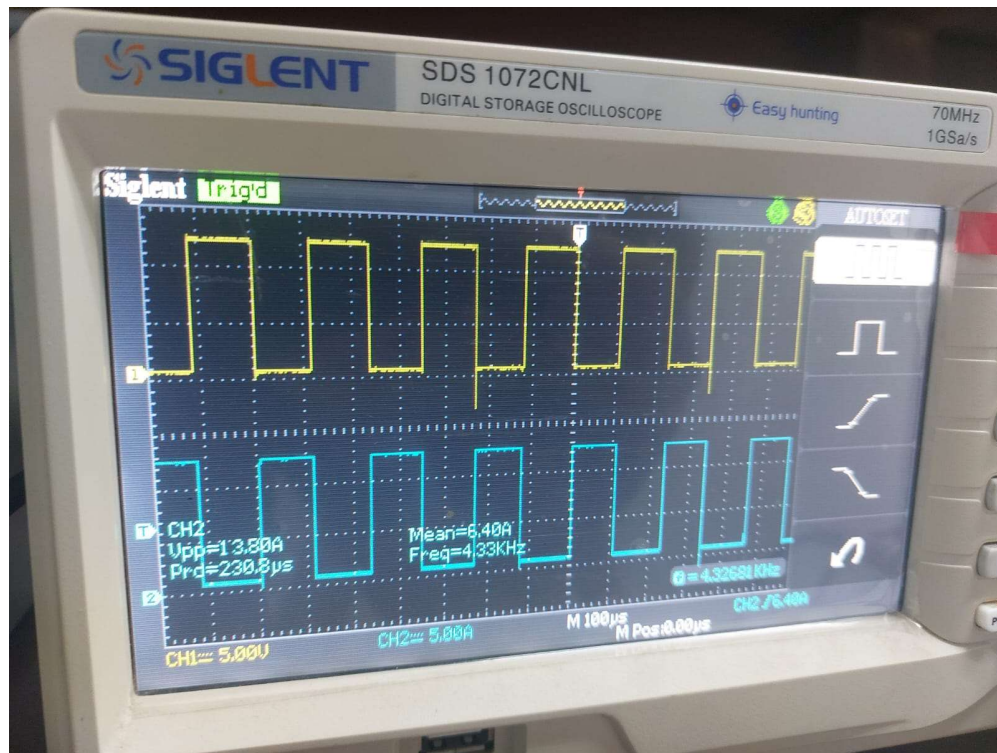
Datasheet: <https://datasheetspdf.com/pdf-file/1493156/LGE/D400/1>

Circuit design



In this design, in case the non-overlapping circuit gives a high (1) output at the same time, which means that if the worst-case scenario happens, both the MOSFETs will be ON at the same time. If so, the circuit will be short circuited. To avoid that in some instant, I used a pull-down resistor between the output of the buffer circuit and ground (This is used for both the MOSFETs).

Output waveform of the buffer circuit



7 Choosing the MOSFET

When choosing the MOSFETs, we need to consider whether the MOSFET can handle the I_{primary} current driven from the transformer.

$$V_{\text{primary}} I_{\text{primary}} = V_{\text{secondary}} I_{\text{secondary}}$$

$$12 \times I_{\text{primary}} = 65 \times 250 \times 10^{-3}$$

$$I_{\text{primary}} = 1.3541\text{A}$$

I used IRF520 power MOSFET(N-Channel) to do this job, as it can handle this primary current.

According to the datasheet, the maximum current rating (I_{D_rms}) = 9.2A

Datasheet of IRF520: <https://www.vishay.com/docs/91017/irf520.pdf>

Why do we need this block for the inverter design? And why do we need 2 MOSFETs?

In the inverter design, the push-pull configuration involving the two additional MOSFETs enhances the performance and efficiency of the circuit. This arrangement allows for a more balanced and controlled operation, reducing distortions in the AC output waveform. The push-pull setup is particularly beneficial for bidirectional signal handling, ensuring that both halves of

the AC cycle are efficiently generated, leading to a smoother and more accurate transformation from DC to AC. Additionally, this configuration helps mitigate issues such as transformer core saturation, contributing to improved overall power delivery capabilities and reliability in the inverter circuit.

Basically, these two MOSFETs are used as current amplifiers.

8 Designing the transformer

The transformer block is essential for stepping up the voltage from the regulated 5V signal to the desired 65V output.

This is how the transformer works:

- **Voltage transformation:**

With a primary voltage (V_{primary}) of 12V, the transformer steps up this voltage to a higher level on the secondary side. The center tap provides a reference point, allowing for the generation of both positive and negative halves of the AC waveform.

- **Turns ratio:**

The turns ratio of the transformer determines the voltage transformation. In this case, with a V_{primary} of 12V and $V_{\text{secondary}}$ of 65V, the turns ratio (N) can be calculated as follows:

Specifications given $\Rightarrow V_{\text{out}}^{\text{rms}} = 65\text{V (20 + 45)}$
 $V_{\text{in}} = 12\text{V}$
$$\frac{V_{\text{in}}}{V_{\text{out}}} = \frac{12}{65} \approx \frac{1}{5} = \frac{N_p}{N_s}$$

Turns ratio = 1:5

I wined the transformer like this,

Turns in the primary side (in one tap) = 20

Turns in the secondary side = 100

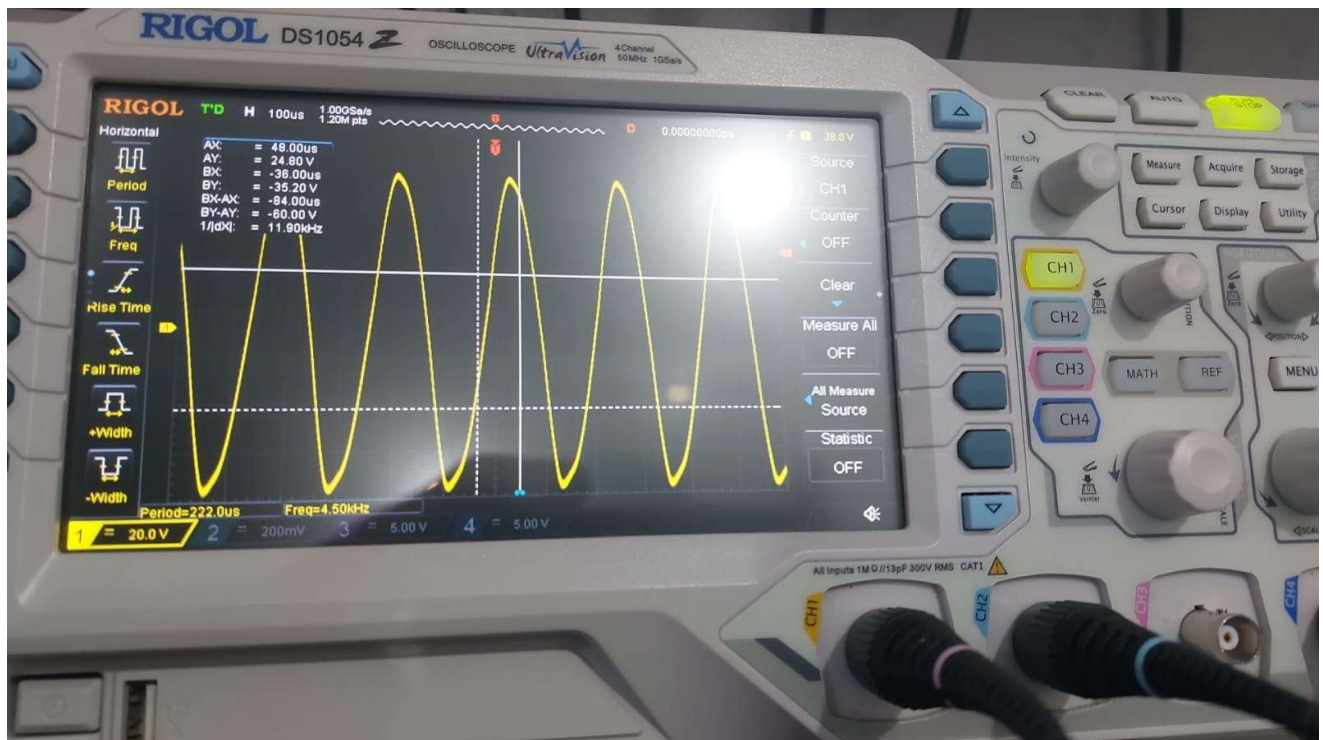
- **AC waveform generation**

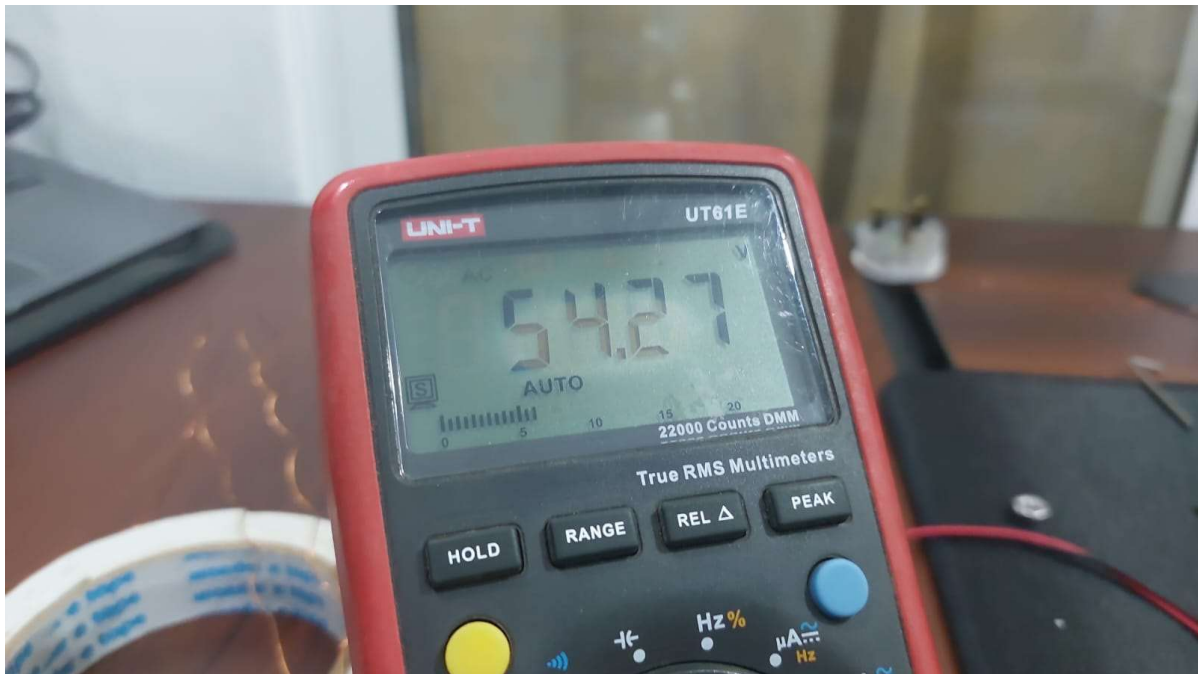
The positive and negative halves of the waveform are produced during alternate cycles because of the center-tapped configuration. The switching mechanism and the transformer's characteristics contribute to the overall frequency and shape of the AC output.

Used materials to make the transformer

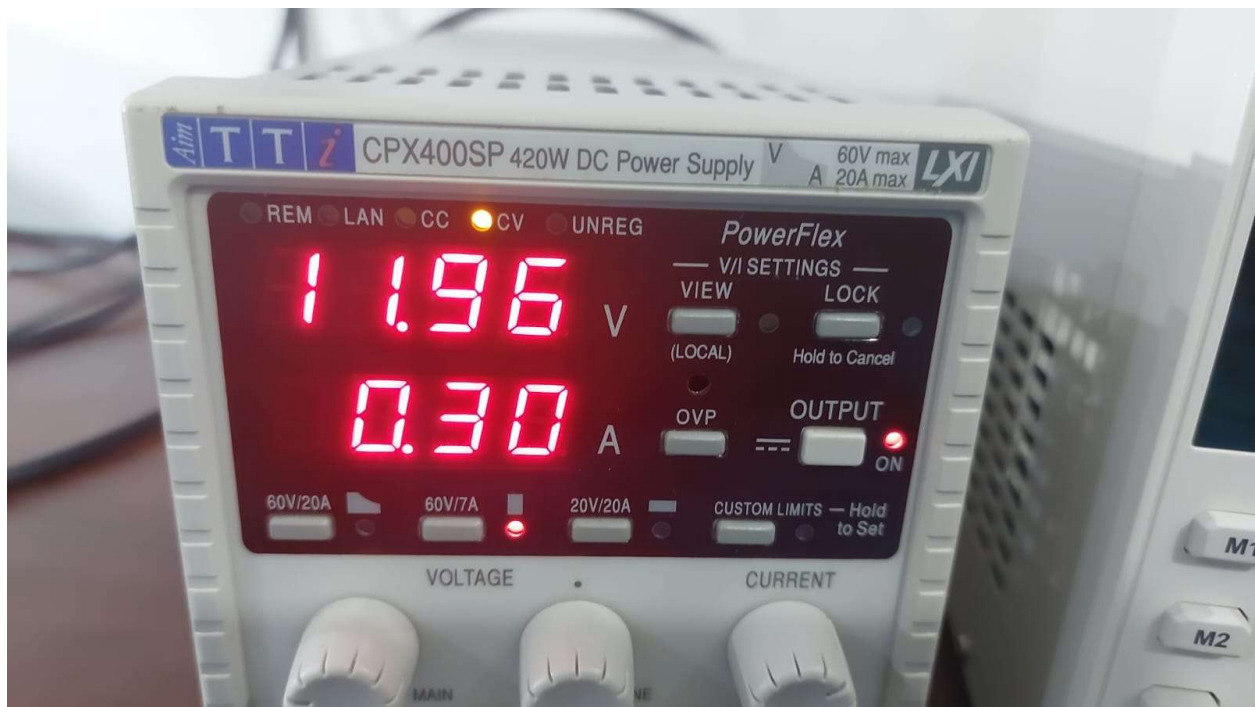
- Bobbin
 - Ferrite core
 - Copper wires
 - Captain tape
- ❖ Used a square shaped bobbin as it is easy to wind smoothly than in a round bobbin.
 - ❖ Used ferrite toroid core to,
 - Get high permeability.
 - Reduce Eddy Current Losses (I laminated the core because of that)
 - Low core losses
 - Get a good high frequency.
 - Used Captain tape to make the windings more stable.

V_{out} waveform of the transformer





$V_{pp} = 54.27V$



I_{out} value	I_{in} value
0 (Open circuit)	23mA
$I_{out}/2 = 26.8\text{mA}$	190mA
$I_{out} = 168\text{mA}$	300mA

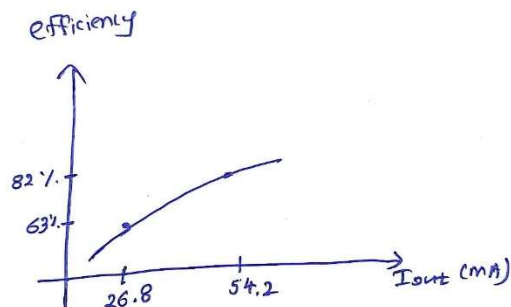
- $I_{out}/2$ is achieved when load = $2\text{k}\Omega$
- I_{out} is achieved when load = $1\text{k}\Omega$

Load resistance is low when the load current is high according to the Ohm's law.

9 Analysis

- Output load = $2\text{k}\Omega$
 - Input current = 190mA
 - Input voltage = 12V
 - Output current = 26.8mA
 - Output voltage = 53.6V
 - $Efficiency = \frac{Output\ current \times Output\ voltage}{Input\ current \times Input\ voltage} = \frac{26.8\text{mA} \times 53.6\text{V}}{190\text{mA} \times 12\text{V}} = 63\%$
- Output load = $1\text{k}\Omega$
 - Input current = 300mA
 - Input voltage = 11.96V
 - Output current = 54.2mA
 - Output voltage = 54.27V
 - $Efficiency = \frac{Output\ current \times Output\ voltage}{Input\ current \times Input\ voltage} = \frac{54.2\text{mA} \times 54.27\text{V}}{300\text{mA} \times 11.96\text{V}} = 82\%$

Efficiency plot



Efficiency increases with output current. Increasing the load will result in increased efficiency.